

- CONSISTENCY OF PMF

$$\sum_{x \in \Omega'} p_x(x) = 1$$

$$\begin{aligned} \sum_{x \in \Omega'} p_x(x) &= \sum_{x \in \Omega'} \mathbb{P}(\{\omega \in \Omega : X(\omega) = x\}) \\ &= \mathbb{P}\left(\bigcup_{x \in \Omega'} \{\omega \in \Omega : X(\omega) = x\}\right) \\ &= \mathbb{P}(\Omega) = 1 \end{aligned}$$

- LINEARITY OF EXPECTATION

$$Z = aX + b$$

$$\Rightarrow E[Z] = a \cdot E[X] + b$$

$$Y = aX + b$$

$$\text{Range of } X: \Omega' = \{x_1, x_2, \dots, x_n\}$$

$$\text{Range of } Y: \Omega'' = \{y_1, y_2, \dots, y_n\}, \quad y_i = a \cdot x_i + b$$

$$p_Y(y_i) = p_X(x_i)$$

$$\begin{aligned} \Rightarrow E[Y] &= \sum_{y \in \Omega''} y_i \cdot p_Y(y_i) \\ &= \sum_{x \in \Omega'} (ax + b) \cdot p_Y(ax + b) \\ &= \sum_{x \in \Omega'} (ax + b) \cdot p_X(x) \\ &= a \cdot \sum x \cdot p_X(x) + b \cdot \sum p_X(x) \\ &= a \cdot E[X] + b \end{aligned}$$

$$p_Y(ax + b) = p_X(x)$$

$$p_X(x)$$

$$* \text{ If } Y = g(X), \text{ then } p_Y(y) = \sum_{\{x: g(x)=y\}} p_X(x)$$

$$!!! \bullet E[Y] = \sum y \cdot p_Y(y) = \sum_y y \cdot \sum_{x: g(x)=y} p_X(x)$$

$$= \sum_y \sum_{x: g(x)=y} y \cdot p_X(x) = \sum_y \sum_{x: g(x)=y} g(x) \cdot p_X(x) = \sum_x g(x) \cdot p_X(x)$$

Law of the Unconscious Statistician

$$\begin{array}{cccc} x: & x_1 & x_2 & x_3 & x_4 \\ g(x): & \searrow & \swarrow & | & | \\ & y_1 & & y_2 & y_3 \end{array} \Rightarrow \sum_y \sum_{x: g(x)=y} = \sum_x$$

- VARIANCE

$$g(X) = (X - E[X])^2$$

square of deviation of X from the mean

$$\therefore \text{Variance} = E[g(X)]$$

$$\text{Var}(X)$$

$$\text{Standard Deviation} = \sqrt{\text{Variance}} = \sqrt{E[g(X)]}$$

$$\sigma_X$$

$$E[(X - E[X])^2] = E[X^2] - (E[X])^2$$

$$\bullet Y = aX + b$$

$$\text{Var}(Y) = a^2 \cdot \text{Var}(X)$$

- INDICATOR RANDOM VARIABLE

$$1_A(\omega) = \begin{cases} 1, & \text{if } \omega \in A \subseteq \Omega \\ 0, & \text{otherwise} \end{cases}$$

$$\Omega = \{1, 0\}$$

$$\begin{array}{cc} \swarrow & \searrow \\ \mathbb{P}(A) & 1 - \mathbb{P}(A) \end{array}$$

$$\therefore \text{PMF } p_{1_A}(x) = \begin{cases} \mathbb{P}(A), & x=1 \\ 1 - \mathbb{P}(A), & x=0 \end{cases}$$

$$\mathbb{P}(A) = E[1_A]$$

(CDF? $E[1_A]$? mean? variance? moments?

- BERNOULLI RANDOM VARIABLE

$$X = \begin{cases} 1, & \text{with probability } p \\ 0, & \text{otherwise} \end{cases}$$

Multi-arm bandit problem?

- BINOMIAL

$$Y = B(n, p)$$

$$P(n=k) = \binom{n}{k} \cdot p^k \cdot (1-p)^{n-k}$$